

B.Tech - Information Technology

IA-2 (Even Sem 2425) | IoT

Smart Door Lock System using IoT

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Introduction

The emergence of smart technology has transformed door lock systems, enhancing their **security, convenience, and usability**. By leveraging the power of the *Internet of Things (IoT)*, smart locks offer features such as remote access, real-time monitoring, and automated control for both residential and commercial properties. This Poster seeks to explain the design and developmental features of a smart door lock that employs advanced technologies to improve security and elevate user convenience.

Objective

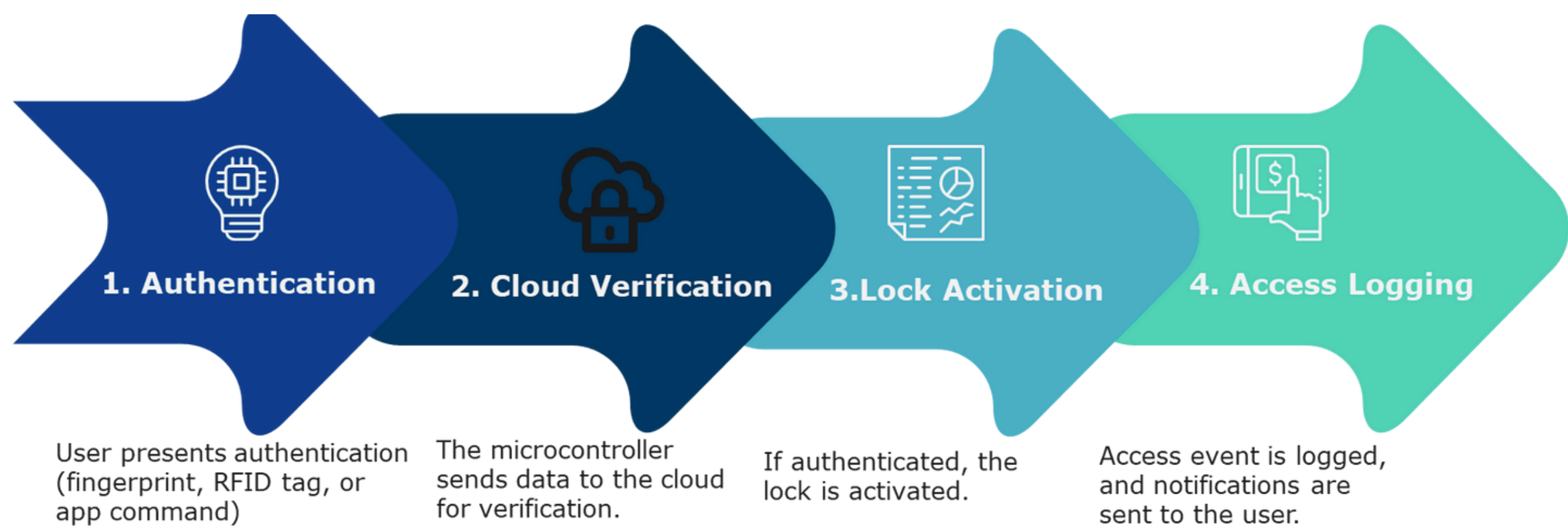
- To design a secure and efficient IoT-based smart door lock system that allows us to:
- Empower yourself with the **convenience of remotely controlling** your door locks via a smartphone
 - Stay secure with **real-time alerts** and access logs that keep you informed of every access attempt
 - User **authentication via RFID, fingerprint, or mobile app**
 - Integration with **cloud services** for data storage and monitoring

System Architecture

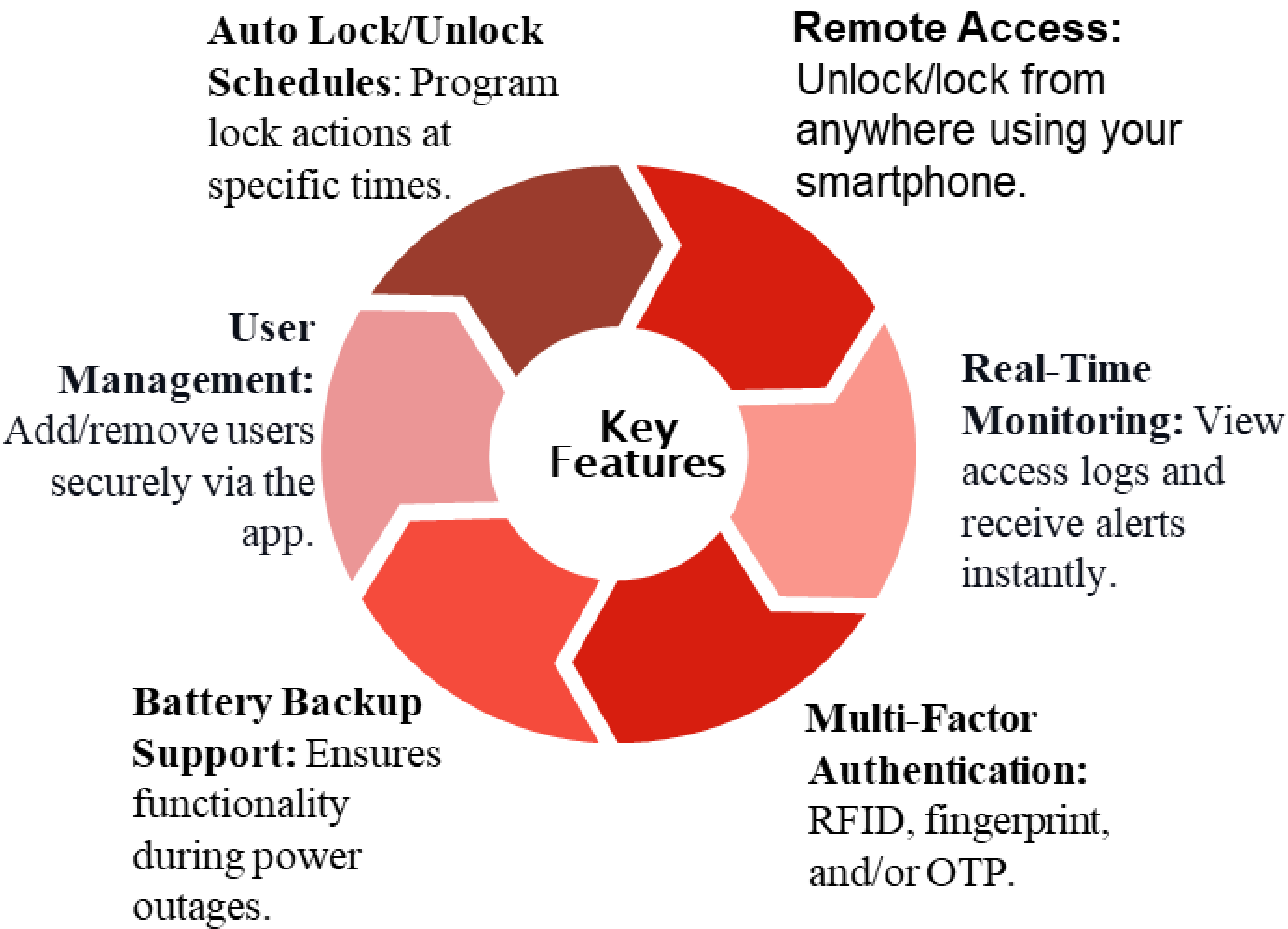
The system uses a microcontroller connected to a locking mechanism and input sensors, all managed via a Wi-Fi-enabled IoT platform.

- Key Elements:**
- ESP8266 NodeMCU:** Controls the lock and connects to Wi-Fi
 - Fingerprint Module/RFID Reader:** Verifies user identity
 - Electromagnetic Lock/Servo Motor:** Physically locks/unlocks the door
 - Wi-Fi Communication:** Sends and receives data to/from the cloud
 - Mobile App or Web Dashboard:** User interface for monitoring and control
 - Firebase/Thingspeak:** Stores access logs and system status in real-time

Secure Entry-Workflow



Key Features



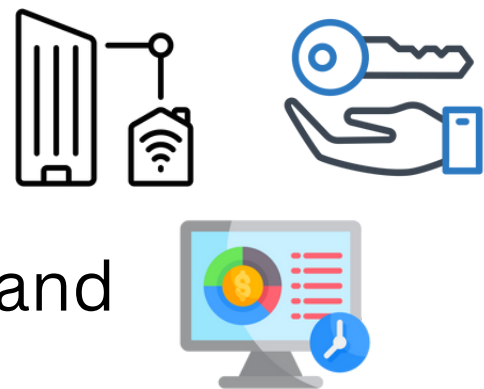
Applications

- Smart homes
- Corporate Offices
- Co-Living Spaces
- Hotel Room Access Systems



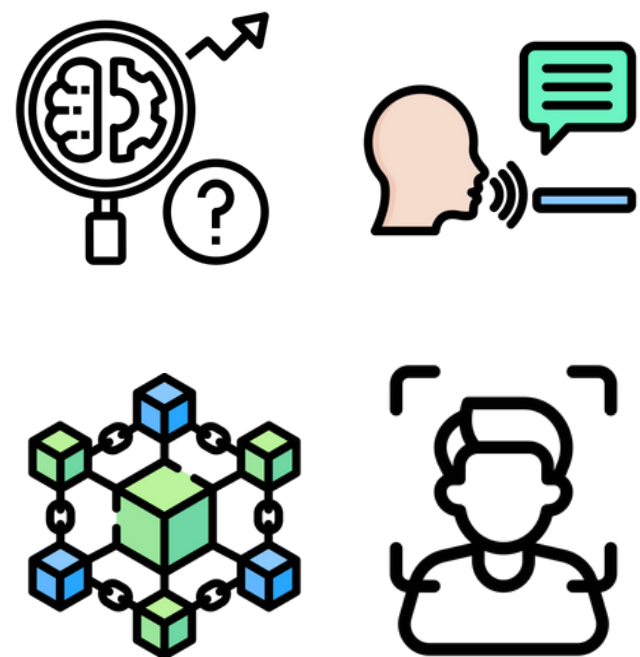
Benefits

- Improved security with real-time tracking
- Eliminates risks associated with lost keys
- Ideal for homes, offices, and institutions
- ease of use with the user-friendly interface and customizable settings.



Future Requirements

- AI-based behavioral access predictions
- Voice assistant integration (Alexa, Google Assistant)
- Blockchain-secured access records
- Facial recognition or gesture-based unlocking



References

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